

Bridging Graph Polynomials and Symmetric Tensors

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Overview

- ▶ Highlight a relationship between symmetric tensors and graph polynomials

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- ▶ Highlight a relationship between symmetric tensors and graph polynomials
- ▶ Investigate the nature of this relationship

Symmetric Tensor Recap

- ▶ *"Symmetric tensors are objects that transform like symmetric tensors"*

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- ▶

Relationship with Homogenous Polynomials

Veronese Variety

Graph Terminology Required

Constructing (particular) Graph Polynomials

Properties

What is... the Grothendieck Ring of Varieties

Belkale, Brosnan (2004)

What is the connection?

Image of R on the Veronese

Localisations in Context

Generating of Symmetric Tensors

Further Questions

Bibliography